

Poisoning the Threshold-Calibration Stage in Federated IoT Anomaly Detection: A Policy-Differentiated Vulnerability Analysis

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Abstract. Federated learning (FL) defenses for IoT anomaly detection largely target training and aggregation, while the post-training *threshold-calibration stage* lets each client autonomously collect benign calibration scores and derive its detection threshold. This paper characterizes that stage under a score-level contamination model: one compromised client replaces a fraction f of its calibration buffer with tail-sampled benign reconstruction scores, shifting the threshold without altering model weights, training data, gradients, aggregation, test scores, or labels. Across Global, Local, and Cluster threshold policies on the N-BaIoT nine-device federation, the raising attack meets a pre-specified material-displacement criterion in every evaluated non-zero policy-fraction condition and reduces the compromised client’s true-positive rate by 2.4–7.9 percentage points (pp), including 4.9–7.9 pp at $f=0.4$. The lowering attack raises cross-client false-positive-rate disparity for Local and Cluster, increasing CV(FPR) by 0.05–0.06 at $f=0.4$ under fixed-budget replacement; the associated rise in true-positive rate reflects threshold displacement, not a security benefit. Policy choice governs both magnitude and threshold propagation: Local confines the shift to the compromised client, Cluster produces smaller non-victim threshold perturbations through cluster recomputation and assignment changes, and Global dilutes the shift by $1/N$ but applies it federation-wide. Non-victim true-positive rates were not recomputed, so effects beyond the compromised client are characterized at the threshold level and, for lowering, at the fleet-FPR level. Bounded to N-BaIoT and score substitution, the results describe a vulnerability mechanism rather than a complete traffic-generation attack.

Keywords: Federated Learning · Anomaly Detection · IoT Security · Poisoning Attack · Threshold Calibration · Vulnerability Analysis

1 Introduction

Federated learning has been widely studied for IoT anomaly detection, allowing devices to contribute to a shared model without centralizing raw traffic [15]. Threshold calibration then determines the operating point used by each detector. Security work on federated anomaly detection has concentrated on the *training phase*: Byzantine clients injecting adversarial gradients, label-flipping poisoning, or model-inversion attacks [4,5,19]. Aggregation-phase defenses such as FLTrust, Krum, and Trimmed-Mean target exactly this surface [6,7,20], and recent FL-IDS work adds personalized, non-IID-robust, knowledge-distillation, and anti-poisoning designs [3,9,14,18], all acting on training data, gradients, model updates, or aggregation inputs.

The step that follows has received much less attention: in the *threshold-calibration stage*, each client collects a window of benign traffic, computes local reconstruction scores, and derives its detection threshold at a fixed quantile q . Training- and aggregation-phase defenses complete before this stage begins, and none of the designs cited above is specified to observe or validate the client-local calibration buffer. If a single client’s benign calibration buffer is contaminated with benign scores from the upper (or lower) tail of that client’s own distribution, its threshold moves.

This paper characterizes that stage under a score-level contamination model in which an adversary controlling one client replaces a fraction f of that client’s calibration buffer with tail-sampled scores from the same client’s benign calibration split. Model weights, training data, labels, gradients, aggregation, test scores, test labels, and the threshold-policy definition all stay fixed, so score ranking is invariant by construction and only the operating point moves. The intervention is therefore not data poisoning relocated to calibration data (Table 1): it acts strictly after model convergence and exploits a trust assumption specific to this stage. Its consequences depend on how the federation turns local thresholds into deployed ones. That last dependence is the research question: does the choice of a federation-wide, per-client, or cluster-averaged threshold change how far one contaminated buffer propagates and how much it reduces detection? The evaluation covers all three policies—Global (federation-wide mean), Local (per-client), and Cluster (cluster-averaged)—with each of the nine physical-device clients of N-BaIoT [13] taken as the compromised client in turn, at four injection fractions.

The paper makes three contributions. First, it formalizes a single-client, calibration-only threat model and states the invariants it preserves, isolating the post-training calibration buffer as the attacked channel. Second, it reports a paired-seed evaluation of threshold raising and threshold lowering across the three threshold policies, each with a matched Random-Benign negative control that keeps the injection budget fixed but removes the tail structure. Third, it characterizes the resulting policy-differentiated behavior: sensitivity at the compromised client, threshold propagation to non-victims, and downstream operating-point harm. All claims are bounded to the nine-device N-BaIoT

setting with autoencoder detectors and FedAvg aggregation; the mechanism operates at the score level, and defense design is out of scope.

The remainder of this paper is organized as follows. Section 2 reviews federated IoT anomaly detection, training-side defenses, and adjacent threshold-calibration work; Section 3 defines the threat model and attack construction; Section 4 describes the protocol, threshold policies, attack parameterization, and statistical analysis; Section 5 presents the clean-baseline and poisoning results; Section 6 discusses findings and limitations; and Section 7 concludes.

2 Background and Related Work

2.1 Federated IoT Anomaly Detection and Training-Side Defenses

Federated intrusion and anomaly detection for IoT has moved beyond plain FedAvg training [3,9,14,15,18]: these studies improve training-time collaboration and robustness metrics, while thresholded autoencoder pipelines still rely on a benign post-training calibration step. The N-BaIoT benchmark [13], which provides labeled benign and Mirai/BASHLITE traffic for nine physical devices, is a widely used FL-IDS evaluation platform.

The defenses developed for this setting act on the training phase. Byzantine-robust rules, including Krum [6], Trimmed-Mean, and Median [20], filter or downweight outlier gradients before aggregation, and FLTrust [7] anchors aggregation to a trusted server dataset. Recent FL-IDS work extends this training-side view with personalized poisoning defenses, knowledge-distillation verifiers, and probe-guided anti-poisoning pipelines under non-IID IoT data [9,14,18]. All of them inspect training data, gradients, model updates, or aggregation inputs; none is specified to observe the later client-local calibration buffer or designed for the post-training calibration channel considered here—a statement about the defenses reviewed here, not a claim that no calibration-stage defense could be built.

2.2 Threshold Calibration and Adjacent Work

Kloft & Laskov [10] study online poisoning in centralized anomaly detection via a continuous training stream, assuming a trusted aggregator or persistent model-update access—both absent from FL’s post-training calibration stage, where calibration data never leaves the client and the pipelines considered here perform no server-side sanitization. FL model poisoning [2] likewise targets the gradient and weight surface. The closest work treats calibration and thresholding as trusted: Laridi et al. [11] propose summary-statistics-based federated thresholding without adversarial contamination, and Sáez-de-Cámara et al. [16] a clustered federated IoT anomaly-detection architecture that assumes a benign calibration pipeline. The Global, Local, and Cluster policies evaluated here, defined in Section 4, follow the partitioning structures of that literature [8,11,17].

Table 1. Positioning relative to adjacent poisoning literature.

Prior work type	Attack surface	What changes	Distinction from this work
Training-data poisoning [5,19]	Training set / labels	Model parameters	Calibration-stage poisoning leaves training data and model unchanged
FL model poisoning [2,4]	Client gradients	Model weights	Needs no gradient access; acts after convergence
Byzantine aggregation [6,20]	Aggregation inputs	Server aggregate	Does not touch aggregation; harm is post-aggregation
Centralized AD poisoning [10]	Central train stream	Decision boundary	No server-side sanitizer in the FL calibration pipelines considered
Threshold / model personalization [8,11,17]	Threshold policy	Per-client/cluster τ	Studies utility, not adversarial contamination
This work	Calibration scores	Threshold τ only	Score-level, post-training, single-client

Table 1 positions this study against those lines of work. Within the reviewed literature, no prior study isolates post-training threshold calibration as the attacked channel while training data, model, aggregation, and test artifacts remain unchanged, and the security implications of the three policies have not been examined under a calibration-stage adversary. This is not a new poisoning taxonomy; the contribution is isolating the channel and measuring policy-dependent victim effects and threshold propagation under a controlled score-level intervention. Section 3 states the adversary model.

3 Threat Model

3.1 Calibration Stage and Adversary Model

Detection proceeds in a fixed sequence: local training, model convergence under FedAvg, benign calibration at each client, threshold construction under the federation’s policy, and deployment-time evaluation (Fig. 1). Benign calibration is the step attacked here. Each client i collects a benign calibration set $\mathcal{C}_i = \{s_1, \dots, s_{n_i}\}$ of reconstruction scores from its own traffic, then sets its local threshold $\tau_i = \text{quantile}(\mathcal{C}_i, q)$ at a fixed quantile q (here $q=0.95$), which the federation’s policy uses directly or averages (Section 4). The buffer is collected autonomously by the client once the aggregation round completes, and the defenses considered here neither monitor nor validate it.

The adversary controls a single compromised client v and can read and modify that client’s own calibration buffer before threshold computation. Its access is otherwise restricted. It cannot modify training data, training labels, model weights, gradients, the aggregation procedure, test scores, test labels, other clients’ data, client eligibility, or the threshold-policy definition, which the federation fixes. The reservoir is drawn exclusively from the same client’s own benign calibration split (Section 4): attack-labeled samples, test scores, and training scores never enter it. This capability differs from the access assumed for model-poisoning adversaries rather than being strictly weaker: the adversary needs no gradient crafting or knowledge of the aggregation rule, but assumes read and write access to the calibration buffer.

Because per-client detection harm is measured at the compromised client, “victim” refers to client v itself and “non-victim clients” to the remaining members. Raising the threshold at v reduces that client’s own detection rate on attack traffic. The contamination also reaches clients the adversary does not control whenever the policy shares thresholds: under Global every member receives the contaminated threshold, under Cluster co-cluster members may receive a perturbed one, and under Local no other client is affected.

3.2 Attack Construction, Controls, and Scope

Both objectives use the same fixed-budget replacement rule (Replace-Fixed-Budget): at $f=0$ no replacement occurs ($m=0$) and the cell is the unpoisoned clean baseline, while for $f>0$ the attack replaces $m = \max(1, \text{round}(fn_v))$ positions of $\mathcal{C}_v$ with scores drawn with replacement from a tail of $\mathcal{C}_v$ itself. Threshold-Raise, the primary objective, draws from the top-10% tail (High-Score-Benign), which increases τ_v and suppresses alarms at the victim site. Threshold-Lower, the secondary and conditional objective, draws from the bottom-10% tail (Low-Score-Benign), which decreases τ_v , raises the false-positive rate, and may increase cross-client FPR disparity depending on the threshold policy; the accompanying $\Delta\text{TPR}>0$ is a mechanical consequence of moving the decision boundary downward and is not a security benefit. The matched negative control, Random-Benign, samples uniformly from the full calibration distribution instead of a tail, holding the injection budget fixed while removing the tail structure; any effect of tail structure, not position perturbation, should vanish there.

Score-level manipulation does not demonstrate realizable raw-traffic generation, which Section 6 takes up as future work; claims are bounded to a single compromised client, and collusion is not evaluated.

4 Experimental Protocol

4.1 Dataset and Federation

The evaluation uses N-BaIoT [13], a labeled IoT traffic dataset covering nine physical devices across five types (doorbell, baby monitor, security camera, thermostat, webcam), each device forming one federation client; all nine satisfy the

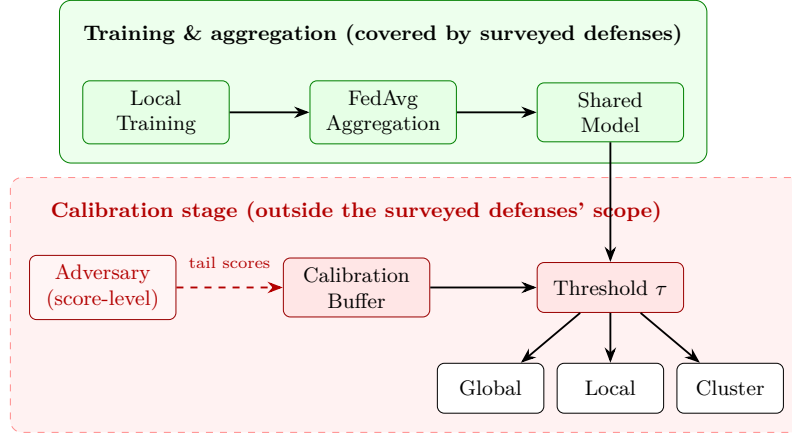


Fig. 1. Attack surface: threshold calibration occurs after FL training. Green boxes mark training and aggregation; the dashed red region marks calibration, outside the scope of the defenses surveyed in Section 2. Score-level access to one client’s calibration buffer can shift τ without modifying model weights, gradients, or labels.

≥ 100 -sample calibration eligibility threshold (size range in Table 2). Each device’s benign stream is partitioned chronologically into a 60% train / 1% gap / 20% calibration / 1% gap / 18% test split, the gaps avoiding temporal leakage, and training and calibration use benign traffic only. Attack-labeled records are scaled with each device’s training-fitted scaler and used only as the positive class for TPR; they never enter training, calibration, poisoning reservoirs, or injected buffers. The federation trains a shared autoencoder on benign traffic using FedAvg [12] (Table 2); each client then scores its own traffic with the aggregated model and enters the threshold-calibration stage at $q=0.95$.

4.2 Threshold Policies

Three threshold-policy families are evaluated, covering the design points surveyed in Section 2.

Global.

$$\tau_i^{\text{local}} = \text{quantile}(\mathcal{C}_i, q), \quad \tau^{\text{global}} = \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{local}}.$$

Every client receives the same τ^{global} , so a single poisoned local threshold contributes only $1/N$ of the shared shift.

Local. Each client retains $\tau_i = \tau_i^{\text{local}}$. Shifts are confined to the compromised client; no cross-client threshold propagation.

Cluster. Clients are grouped on a standard-scaled 4-feature fingerprint (mean, std, skew, p_{95} of $\mathcal{C}_i$) via k -means++ [1] ($K=3$, random seed 42; clustering recomputed after each attack draw so a poisoned fingerprint can change the compromised client’s cluster assignment). The shared threshold is the

Table 2. Experimental configuration: architecture, federated training hyperparameters, and calibration settings.

Component	Configuration
<i>Autoencoder architecture</i>	
Input features	115 (N-BaIoT statistical traffic features)
Encoder layers	115 $\rightarrow$ 80 $\rightarrow$ 40 $\rightarrow$ 20; decoder mirrors
Activation	ReLU (no batch normalization)
Loss	Mean squared reconstruction error
Optimizer	Adam, learning rate 0.001, weight decay 0
<i>Federated training (FedAvg)</i>	
Local epochs	1
Aggregation	Sample-count-weighted mean
Rounds	40 initial, 150 max (conv.: $\Delta\text{loss} \leq 0.005$ over 10)
Batch size	256
<i>Calibration (per-device sizes in released split manifest)</i>	
Benign samples / device	2,622 (Ecobee) – 35,048 (Philips)
Threshold quantile q	0.95

cluster mean, so a compromised client may perturb the threshold for co-cluster members (*intra-cluster threshold perturbation*) while other clusters are unaffected. Under Threshold-Raise, tail injection raises the mean, p_{95} , and dispersion components of the fingerprint, displacing the victim, in the runs observed here, into an effectively singleton cluster whose mean collapses to the victim’s poisoned local threshold; this yields near-Local victim shifts while co-cluster members see perturbations small relative to the victim shift but not identically zero.

4.3 Attack Parameterization

For each (policy, objective, injection fraction $f \in \{0.0, 0.1, 0.2, 0.4\}$) combination the protocol runs 10 seeds, each fixing a training configuration and an independent poisoning draw; a clean copy of the calibration array is preserved and only the poisoned copy is used for threshold computation. The four source-objective pairs are High-Score-Benign + Threshold-Raise (primary), Low-Score-Benign + Threshold-Lower, and Random-Benign with each objective as matched negative controls. The attack is applied to each of the nine clients as victim in turn (single-client scope), yielding 4 source-objective pairs $\times$ 3 threshold policies $\times$ 4 injection fractions $\times$ 9 victim clients $\times$ 10 seeds = 4,320 evaluated cells, 3,240 of them with non-zero injection; the 1,080 cells at $f=0$ are the clean comparators and coincide across source-objective pairs. Training seed i is combined with poisoning seed $i+100$ and analysis seed $i+300$, separating training, calibration-sampling, and resampling stochasticity.

4.4 Statistical Analysis

Clean and poisoned conditions are paired: for a policy, victim v , and seed s , the clean comparator is the $f=0$ cell produced by the same trained model and the same calibration array before replacement, so every reported difference is within-seed and within-victim,

$$\Delta\tau_{v,s} = \tau_{v,s}^{\text{poisoned}} - \tau_{v,s}^{\text{clean}},$$

and likewise for ΔTPR and $\Delta\text{CV}(\text{FPR})$. The nine victim values within a seed are averaged into one seed-level aggregate, and the ten paired aggregates are the inferential units: bootstrap resampling is over those ten seeds, never over the ninety victim–seed cells, which share a trained model within a seed and are descriptive support, not independent runs. Three gates, all fixed before any poisoned run, structure the claims below.

Gate 1 (material shift, primary gate) evaluates the signed displacement $\Delta\tau_{v,s}$ at the victim–seed cell. A cell is *material* when $|\Delta\tau_{v,s}| > \delta\tau_{v,s}$ and its sign matches the objective—positive for raising, negative for lowering—so displacement in the wrong direction never counts, where

$$\delta\tau_{v,s} = \max(0.1 \text{ IQR}(\mathcal{C}_{v,s}^{\text{clean}}), 0.01 \text{ median}_j(\text{IQR}(\mathcal{C}_{j,s}^{\text{clean}})))$$

is computed from the same seed’s clean calibration arrays with j over all nine clients. A *seed passes* when at least 5 of the 9 clients, each taken as victim in turn, are material, and a (policy, fraction) configuration *passes Gate 1* when at least 8 of the 10 seeds pass; these majorities are conservative consistency requirements, not calibrated error rates. Passing supports the claim that the channel displaces thresholds by more than clean-baseline instability at a majority of victims and reproducibly across seeds; failure would mean the displacement is not separable from that instability. Both coefficients were locked from clean-artifact statistics beforehand, the 10% IQR term exceeding the threshold instability observed across clean seeds and the 1% cross-client floor stopping clients with near-zero score spread from counting as material on arbitrarily small shifts. Gate 1 is thus a pre-committed materiality floor rather than a significance test; its pass rates are reported alongside the effect sizes in Table 4.

Gate 2 (directional excess over control) is an objective-matched paired contrast: within each cell the control $\Delta\tau$ is subtracted from the attack $\Delta\tau$, sign-oriented by objective, and the nine victim values are averaged to one value per seed. A configuration passes when at least 8 of the 10 seed values are strictly positive and their median is positive, with a percentile bootstrap CI on those ten contrast values computed alongside but not entering the rule.

Gate 3 (downstream operating-point displacement) evaluates the mean victim ΔTPR at the seed level and passes when its direction matches the threshold objective—negative for raising, positive for lowering—in at least 8 of the 10 seeds; failure means the operating point did not move consistently with the threshold shift. For raising, the threshold increase means reduced victim TPR and hence missed-detection harm. For lowering, the threshold decrease means

a higher false-positive rate and, depending on policy, greater cross-client FPR disparity, and the accompanying positive ΔTPR is evidence of operating-point displacement only; the harm metric for that objective is $\Delta\text{CV}(\text{FPR})$.

All CIs are 95% percentile bootstrap over the 10 seed-level aggregates with $B=10,000$ resamples, reported per configuration, two-sided and without multiplicity adjustment; at ten paired seeds they are approximate uncertainty estimates, not exact coverage guarantees.

4.5 Isolation Sanity Check

To confirm that the attack is confined to the calibration stage, the protocol verifies that (i) AUROC is invariant under poisoning, (ii) per-client rank ordering of clean anomaly scores is unchanged, and (iii) model weights before and after calibration are bitwise identical; all three hold in every cell, including the 3,240 with non-zero injection. Being invariant by construction, these quantities verify implementation isolation rather than provide independent evidence.

5 Results

5.1 Clean-Baseline Policy Profile

Table 3 reports clean-baseline dispersion across all ten seeds; lower $\text{CV}(\text{FPR})$ means more uniform per-client false-positive treatment, and Local is lowest (0.333 ± 0.034), Global highest (0.905 ± 0.039), Cluster intermediate (0.615 ± 0.143) with cluster-assignment sensitivity. Pairwise gaps, taken as within-seed paired differences, are positive with bootstrap CIs excluding zero: G–L 0.572 [0.537, 0.607], G–C 0.291 [0.214, 0.358], C–L 0.281 [0.201, 0.374] (unrounded per-seed aggregates); Fig. 2 shows the per-client FPR distributions. Lower dispersion does not imply better detection: Global retains the best P10 Macro-F1, effectively a worst-client value with nine clients, at 0.398 against Local’s 0.335.

Table 3. Clean baseline policy profile on N-BaIoT ($n=10$ seeds, 9 clients). CV: coefficient of variation; BA: balanced accuracy; P10 F1: P10 Macro-F1; FPR/TPR: client/seed means; $\pm$ is the standard deviation across the ten seeds, not a confidence interval; bold marks the best point estimate, untested for significance; * std ≈ 0.0002 .

Policy	CV(FPR)	CV(TPR)	Worst BA	P10 F1	FPR	TPR
Global	0.905 $\pm$ 0.039	0.255$\pm$0.030	0.657 $\pm$ 0.008	0.398 $\pm$ 0.053	0.041	0.793
Local	0.333$\pm$0.034	0.296 $\pm$ 0.036	0.655 $\pm$ 0.005	0.335 $\pm$ 0.058	0.043	0.777
Cluster	0.615 $\pm$ 0.143	0.313 $\pm$ 0.009	0.656 $\pm$ 0.004	0.299 $\pm$ 0.000*	0.043	0.742

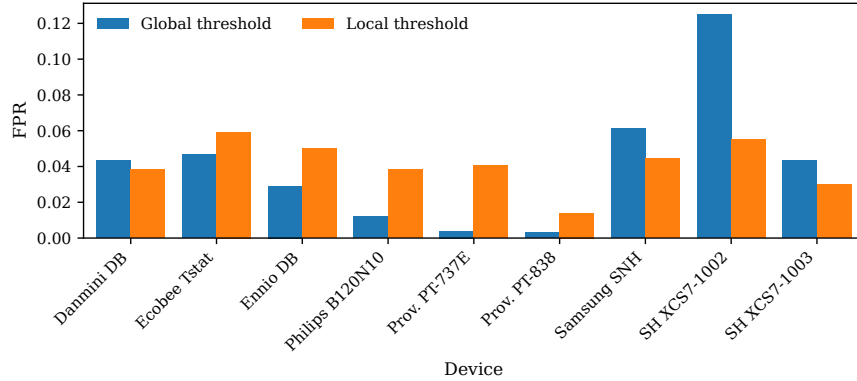


Fig. 2. Clean-baseline FPR heterogeneity is larger under the shared Global threshold than under Local calibration (seed 0). All-seed dispersion, including Cluster, is in Table 3. Device labels: DB, doorbell; Tstat, thermostat; SH, SimpleHome; Prov., Provision.

5.2 Raising Attack: Primary Results

Table 4 and Fig. 3(a) report threshold shift and downstream detection harm for High-Score-Benign + Threshold-Raise.

Under raising, all three policies clear Gates 1 and 3 at every non-zero fraction, with a Gate-1 pass rate of 10/10 seeds throughout, a seed passing when at least 5 of its 9 victim clients show a material, sign-correct shift. For Local and Cluster all nine clients are individually material in every seed; for Global at $f=0.1$ two Provision security cameras (PT-737E and PT-838) are material in only 2/10 and 4/10 seeds respectively and one thermostat in 9/10, yet the victim-majority condition still holds because at least six of the nine are material in every seed. The primary and control $\Delta\tau$ intervals are disjoint everywhere, though a control interval covering zero does not imply an exactly zero control effect. For the control, mean $\Delta\tau$, averaged over the two objectives, is two orders of magnitude smaller than the tail-injection shift at the same fraction, no seed meets the Gate-1 criterion (0/10), and $|\Delta\text{TPR}| \leq 1$ pp at every fraction and policy (Table 4 gives representative $f=0.4$ rows). That magnitude gap is the basis for attributing the targeted effects to tail contamination rather than position perturbation. The pooled control rows provide a compact descriptive summary; Gate 2 is evaluated using the objective-matched Random-Benign conditions separately.

Magnitude differs by policy: at $f=0.4$, Local and Cluster give the largest shifts, $\Delta\tau = +0.902 [+0.864, +0.944]$ and $+0.899 [+0.859, +0.941]$, while Global gives a diluted but still material $+0.100 [+0.096, +0.105]$. The $1/N$ dilution is definitional, yet the residual ($\approx 10\%$ of the local shift) still produces measurable harm.

Mean victim ΔTPR is negative and directionally consistent across all ten seeds for all three policies, so Gate 3 passes everywhere. At $f=0.4$: Local -7.9 pp, Cluster -4.9 pp, Global -6.1 pp. Cluster equals its $f=0.2$ estimate to one deci-

Table 4. Calibration-stage poisoning shifts thresholds under all policies (N-BaIoT, $n=10$ seeds). $\Delta\tau$: mean shift [95% CI]; G1: Gate-1 seed pass rate; ΔTPR : victim detection-rate change. Lowering rows ($\dagger$) report $\Delta\text{CV}(\text{FPR})$, an FPR-disparity metric, as a seed-level mean with no bootstrap interval estimated; Neg. Control: Random-Benign at $f=0.4$, pooled over both objectives; bold=zero G1.

Policy	Attack	f	$\Delta\tau$ [95% CI]	G1	ΔTPR (pp)	$\Delta\text{CV}(\text{FPR})$
Global	Raise	0.1	+0.033 [+0.031, +0.035]	100%	−2.4	—
		0.2	+0.062 [+0.059, +0.065]	100%	−4.6	—
		0.4	+0.100 [+0.096, +0.105]	100%	−6.1	—
Local	Raise	0.1	+0.298 [+0.281, +0.314]	100%	−6.0	—
		0.2	+0.558 [+0.535, +0.581]	100%	−7.3	—
		0.4	+0.902 [+0.864, +0.944]	100%	−7.9	—
Cluster	Raise	0.1	+0.253 [+0.231, +0.277]	100%	−3.9	—
		0.2	+0.504 [+0.475, +0.532]	100%	−4.9	—
		0.4	+0.899 [+0.859, +0.941]	100%	−4.9	—
Global	Lower $\dagger$	0.1	−0.006 [−0.006, −0.006]	60%	+0.8	−0.007
		0.2	−0.010 [−0.011, −0.010]	100%	+1.1	−0.011
		0.4	−0.018 [−0.020, −0.017]	100%	+2.1	−0.019
Local	Lower $\dagger$	0.1	−0.053 [−0.057, −0.049]	100%	+2.8	−0.002
		0.2	−0.094 [−0.100, −0.088]	100%	+5.9	+0.005
		0.4	−0.164 [−0.176, −0.154]	100%	+11.8	+0.053
Cluster	Lower $\dagger$	0.1	−0.047 [−0.065, −0.031]	100%	+2.2	+0.007
		0.2	−0.100 [−0.115, −0.084]	100%	+5.4	+0.011
		0.4	−0.154 [−0.167, −0.142]	100%	+10.3	+0.060
Global	Neg. Ctrl	0.4	−0.000 [−0.001, +0.001]	0%	+0.3	—
Local	Neg. Ctrl	0.4	−0.002 [−0.009, +0.006]	0%	+0.9	—
Cluster	Neg. Ctrl	0.4	−0.009 [−0.022, +0.003]	0%	+0.4	—

mal, but no interval on that difference was estimated, so this apparent saturation is not separated from seed-level variation. Downstream harm need not track $\Delta\tau$ monotonically, consistent with each policy’s clean threshold sitting at a different point of the victim’s score distribution; that distribution was not characterized here, so the explanation remains an interpretation.

5.3 Lowering Attack: Secondary Results (Conditional)

Fig. 3(b) shows threshold shift for Low-Score-Benign + Threshold-Lower, whose goal is to increase false-alarm burden; the $\Delta\text{TPR}>0$ below is a threshold-displacement indicator, not a security benefit. At $f=0.4$, Local reaches $\Delta\tau = -0.164$ [−0.176, −0.154] with $\Delta\text{TPR} = +11.8$ pp. Cluster reaches -0.154 [−0.167, −0.142] with +10.3 pp. Global is structurally bounded: $\Delta\tau = -0.018$ [−0.020, −0.017], with $\Delta\text{TPR} = +2.1$ pp, and its Gate 1 passes only at $f \geq 0.2$.

The disparity metric $\Delta\text{CV}(\text{FPR})$ (dimensionless) is +0.053 for Local (positive in 10 of 10 seeds) and +0.060 for Cluster (positive in 8 of 10 seeds), while Global shows −0.019. Under a shared threshold every client’s FPR moves in the same direction, which can reduce the coefficient of variation even as absolute

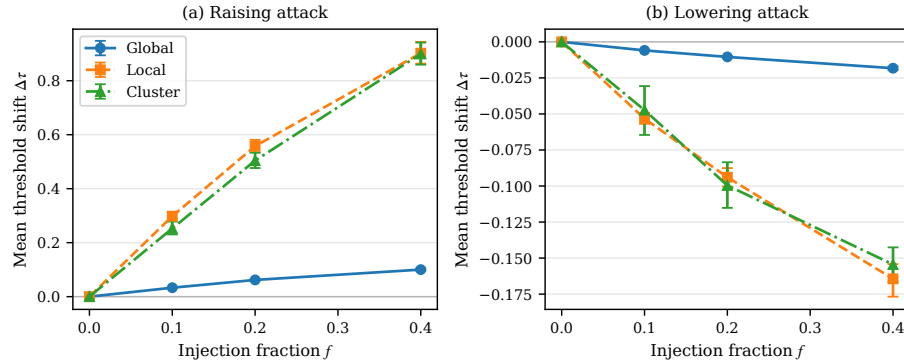


Fig. 3. Mean threshold shift $\Delta\tau$ vs. injection fraction f for raising (a) and lowering (b) attacks. Error bars are 95% bootstrap CIs over 10 seeds; Global is diluted by $1/N$, and Random-Benign controls are near-null (Table 4). Panel (b) uses the legend of panel (a).

FPR rises; that negative value reflects dispersion reduction, not reduced alarm burden. Absolute clean $\rightarrow$ poisoned FPR at $f=0.4$: Global $0.041 \rightarrow 0.043$, Local $0.043 \rightarrow 0.046$, Cluster $0.043 \rightarrow 0.049$. Lowering a $q=0.95$ quantile requires displacing scores in the upper-tail support, which random position replacement achieves with probability $\approx f$, leaving 60% of that support intact at $f=0.4$; Global compounds the effect via $1/N$ dilution, so Replace-Fixed-Budget may understate the displacement achievable by a rank-aware replacement strategy.

5.4 Policy-Differentiation Summary

Table 5. Global propagates diluted threshold shifts federation-wide, while Local confines them to the victim ($f=0.4$, $n=10$ seeds). Material non-vic.: mean number of the eight non-victims, over ten seeds, whose $\Delta\tau$ meets the Gate-1 criterion; it measures threshold materiality, not downstream harm. [†] Shared threshold; [‡] cluster recomputation and assignment changes; std = 0.017 across the 90 victim-seed cells.

Policy	Victim $\Delta\tau$	Victim ΔTPR (pp)	Non-vic. $\Delta\tau$	Material non-vic.
Global	+0.100	-6.1	+0.100 [†]	$\approx 8/8$
Cluster	+0.899	-4.9	+0.0003 [‡]	$\approx 6/8$
Local	+0.902	-7.9	0.000	0/8

Table 5 quantifies threshold propagation under the raising attack at $f=0.4$. Global applies the same shift to all nine clients, so non-victim $\Delta\tau$ equals victim $\Delta\tau$ by construction; the material non-victim count nonetheless falls below 8/8 in a minority of seeds, because the Gate-1 floor $\delta\tau_{j,s}$ scales with each client’s calibration IQR and a shared shift can fall below that floor at the widest-spread clients.

Cluster produces near-Local victim shifts (+0.899 vs. +0.902) with smaller victim ΔTPR (−4.9 vs. −7.9 pp), and its non-victim changes include positive and negative values (mean +0.0003; std 0.017 across the 90 victim-seed cells, arising from cluster recomputation and assignment changes rather than aggregation). The small mean therefore reflects cancellation, not negligible perturbation: on average 6 of the 8 non-victims meet the Gate-1 criterion, at magnitudes about an order of magnitude below Cluster’s weakest victim shift (+0.253 at $f=0.1$). Cluster non-victim true-positive rates were not recomputed. Local yields the highest per-victim shift with no material non-victim shifts at all, by policy design.

6 Discussion and Limitations

The experiment operates at the score level, replacing calibration-buffer scores programmatically rather than through generated traffic. A complete end-to-end attack would also require generating packets whose reconstruction error lands in the target tail, which means controlling feature-level statistics through the encoder-decoder nonlinearity under real network behavior; this work is a *vulnerability-surface characterization*, not a deployed attack. With-replacement sampling from the tail reservoir at $f=0.4$ also leaves duplicate scores in the buffer—on the order of 10^3 expected collision pairs for the smallest calibration set ($n_v \approx 2,622$)—which a uniqueness check would flag. Perturbing scores within the tail removes the duplicates, at the cost of leaving the exact tail boundary, so the artifact belongs to the sampling rule rather than the attack; detecting a buffer contaminated without it remains open.

The Local and Cluster lowering results are *conservative under the evaluated replacement rule*: an adversary exploiting the buffer’s rank structure may achieve larger displacements and alarm-burden effects, which was not evaluated here. Within the evaluated setting the results are consistent with a trade-off in policy design: Local maximizes per-client isolation at the cost of the largest per-victim exposure, whereas Global limits per-victim shift by averaging but applies residual shifts federation-wide.

Claims are bounded to N-BaIoT (9 clients), the autoencoder architecture, FedAvg, and $q=0.95$; generalization to other datasets, detectors, FL algorithms, larger federations, and quantiles is open, as is multi-client collusion. Non-victim true-positive rates were not recomputed, so federation-wide impact under Global and intra-cluster impact under Cluster are characterized at the threshold and fleet-FPR level only, and interval precision is bounded by the ten-seed budget (Section 4). Defense design for the calibration channel—threshold validation, calibration-set integrity constraints—is out of scope and merits independent study.

7 Conclusion

This paper characterized the post-training threshold-calibration stage in federated IoT anomaly detection as an attack surface the studied training- and

aggregation-phase defenses do not address. Under a score-level contamination model on the N-BaIoT nine-device federation (FedAvg autoencoder, $q=0.95$, single compromised client), the raising attack satisfied the pre-specified material threshold-displacement criterion in every evaluated non-zero policy-fraction condition, with 95% bootstrap intervals on $\Delta\tau$ excluding zero, and degraded victim detection rates by 2.4–7.9 pp, consistently signed in all ten seeds. The lowering attack shifts thresholds downward, raising the FPR-disparity metric $\Delta\text{CV}(\text{FPR})$ by 0.05–0.06 for Local and Cluster at $f=0.4$, a conservative result under Replace-Fixed-Budget; the accompanying ΔTPR increase reflects operating-point displacement, not a security benefit. The policy ordering—the largest evaluated per-victim exposure under Local, diluted but federation-wide threshold propagation under Global, smaller non-victim threshold perturbations under Cluster—suggests that in settings resembling this one, calibration-stage hardening should be matched to the threshold policy rather than left to training-phase defenses. These findings isolate a calibration-channel vulnerability mechanism, not a complete traffic-generation attack; traffic-level realizability, calibration-stage integrity defenses, and generalization across detectors and datasets remain open.

Reproducibility. The artifact bundle archived on Zenodo at doi:10.5281/zenodo.21196568 contains the configuration, paired seeds, split manifest, per-cell results, negative-control records, isolation checks, and figure/table scripts. Checkpoints, hyperparameters, and analysis gates were fixed before poisoned runs; no poisoned metric informed any of those choices.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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